9B Lesson 4

Plant products: starch, lipids and proteins

Year 9 Science · 9B Plant growth · Student Book pages 26 to 27

Name _____ Class _____ Date _____

Today I can:

- ☐ I can describe how plants use glucose to make starch, cellulose, lipids and proteins, and explain why nitrates are needed.
- ☐ I can test a leaf for starch and explain what happens when a seed germinates.

Key words

starch a polymer made of many glucose molecules joined together; a plant's energy store

polymer a long molecule made of many small molecules joined in a chain

cellulose a polymer of glucose used to make plant cell walls

lipid a fat or oil; insoluble in water

protein a polymer made of amino acids; enzymes are proteins

amino acid a small molecule that proteins are built from; needs nitrogen

enzyme a protein that speeds up a reaction in a living thing

nitrate a mineral salt that supplies nitrogen for making amino acids

germination when a seed starts to grow into a seedling

iodine solution an orange-brown liquid that turns blue-black with starch

Benedict's solution a blue liquid that turns orange-red when heated with glucose

● Core Everyone does this section.

1. Practical: testing a variegated leaf for starch.

a) Draw the leaf before the test and after adding iodine. Label the green and white parts on the first drawing, and the blue-black parts on the second.

Before the test	After adding iodine

b) Name the solution used to test for starch and the colour it turns when starch is present.

c) Explain why only some parts of the leaf changed colour.

2. Complete the sentences. Use the word bank.

Glucose molecules made in photosynthesis are joined together to make a polymer called _____.

Starch is an energy store. It is found in the chloroplasts, in storage organs such as _____, and in seeds.

Sugars are also used to make _____, which builds plant cell walls, and _____ such as the oils in seeds.

3. Match each substance to one of its uses in a plant. Draw a line.

starch	speed up reactions such as photosynthesis and respiration
cellulose	energy store in seeds and potatoes
lipids	build cell walls
proteins	built from amino acids using nitrogen from nitrates
enzymes	make the waterproof cuticle and cell membranes

●● Stretch When Core is finished.

S1. Seeds contain a store of starch. Describe what happens when a seed germinates. Use the words water, enzymes, glucose, respiration and warm.

S2. Describe a test that would show that glucose is produced in a germinating seed. Include the colour change and one safety precaution.

S3. Explain why plants do not grow well when there is a lack of nitrates in the soil.

●●● Challenge Explain in full sentences.

C1. A plant was left in the dark for 24 hours. A metal foil stencil was then attached to one leaf and the plant was left in bright light for a day. The leaf was then tested for starch.

a) Explain why the plant was left in the dark first.

b) Predict the result of the starch test and explain it.

c) Explain why a metal foil stencil was used and not a clear plastic one.

C2. Avocado fruit flesh is rich in lipids. Avocado seeds also contain lipids. Explain why the plant puts lipids in each place. The two reasons are different.

My thinking (any language)

A potato is a store of starch made by the plant in summer. When does the potato plant use this store, and why does it store starch rather than glucose?

Word bank starch · polymer · cellulose · lipids · potatoes · protein · amino acid · enzyme · nitrate · germination · iodine solution

✂ ---------- cut here

Exit ticket 9B Lesson 4: Plant products: starch, lipids and proteins Name _____

1. Which solution tests for starch, and what colour does it turn?

2. Why do seeds need a store of starch?

3. What are enzymes made of, and which mineral salt do plants need to make them?

Today I can: Goal 1 ☐ Goal 2 ☐